

Project 1: Job Market ETL Pipeline

Project Overview

The Job Market ETL Pipeline is a Data Engineering project designed to extract, transform, load, and analyze job market data. The project processes over 158,000 job records and generates insights using PostgreSQL, SQL, and Power BI.

Objective

To build an end-to-end ETL pipeline that:

- Extracts job market data from CSV files.
 - Cleans and transforms raw data.
 - Loads processed data into PostgreSQL.
 - Performs SQL-based analysis.
 - Creates an interactive Power BI dashboard.
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Technologies Used

Programming

- Python
- Pandas

Database

- PostgreSQL
- DBeaver

Visualization

- Power BI

Other Tools

- Git
 - GitHub
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ETL Architecture

CSV Dataset ↓ Python ↓ Data Cleaning & Transformation ↓ PostgreSQL ↓ SQL Analysis ↓ Power BI Dashboard

Dataset Information

Dataset Name: Job Descriptions Dataset

Total Records: 158,000+

Columns:

- Job Id
 - Experience
 - Qualifications
 - Salary Range
 - Country
 - Work Type
 - Company Size
 - Job Posting Date
 - Job Title
 - Role
 - Skills
 - Company
 - Company Profile
 - salary_min
 - salary_max
 - salary_avg
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Extract Phase

Tasks Performed

- Read CSV file using Pandas.
- Verified dataset structure.
- Checked data types.
- Checked missing values.
- Checked duplicate records.

Example:

```
df = pd.read_csv("job_descriptions.csv")
```

Transform Phase

Missing Value Handling

Company Profile column contained missing values.

Solution:

```
df["Company Profile"] = df["Company Profile"].fillna("Not Available")
```

Duplicate Detection

```
df.duplicated().sum()
```

Duplicates removed using:

```
df.drop_duplicates(inplace=True)
```

Salary Transformation

Created:

- salary_min
- salary_max
- salary_avg

Example:

\$44K-\$88K

became:

salary_min = 44000

salary_max = 88000

salary_avg = 66000

Date Transformation

Converted Job Posting Date to datetime format.

```
df["Job Posting Date"] = pd.to_datetime(  
    df["Job Posting Date"],  
    format="%d-%m-%Y"  
)
```

Country Analysis

Performed country-level job distribution analysis.

Work Type Analysis

Analyzed:

- Remote Jobs
 - Hybrid Jobs
 - Onsite Jobs
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Skills Analysis

Created skill frequency analysis using predefined skill dictionary.

Skills analyzed:

- Python
 - SQL
 - AWS
 - Spark
 - Power BI
 - Tableau
 - Java
 - JavaScript
 - HTML
 - CSS
 - React
 - Angular
 - Docker
 - Kubernetes
 - MongoDB
 - MySQL
 - Git
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Load Phase

Data loaded into PostgreSQL.

```
df.to_sql(  
    "jobs",  
    engine,  
    if_exists="replace",
```

```
index=False  
)
```

SQL Analysis Queries

Top Countries by Job Openings

```
SELECT "Country",  
COUNT(*) AS total_jobs  
FROM jobs  
GROUP BY "Country"  
ORDER BY total_jobs DESC  
LIMIT 10;
```

Work Type Distribution

```
SELECT "Work Type",  
COUNT(*) AS total_jobs  
FROM jobs  
GROUP BY "Work Type";
```

Top Job Roles

```
SELECT "Role",  
COUNT(*) AS total_jobs  
FROM jobs  
GROUP BY "Role"  
ORDER BY total_jobs DESC  
LIMIT 10;
```

Top Companies

```
SELECT "Company",  
COUNT(*) AS total_jobs  
FROM jobs  
GROUP BY "Company"
```

```
ORDER BY total_jobs DESC  
LIMIT 10;
```

Average Salary

```
SELECT AVG(salary_avg)  
FROM jobs;
```

Average Salary by Country

```
SELECT "Country",  
AVG(salary_avg) AS avg_salary  
FROM jobs  
GROUP BY "Country"  
ORDER BY avg_salary DESC;
```

Highest Paying Roles

```
SELECT "Role",  
AVG(salary_avg) AS avg_salary  
FROM jobs  
GROUP BY "Role"  
ORDER BY avg_salary DESC  
LIMIT 10;
```

Power BI Dashboard

Dashboard Components:

KPI Cards

- Total Jobs
- Total Companies
- Total Countries
- Average Salary

Visualizations

- Top Countries by Job Openings

- Work Type Distribution
- Top Job Roles
- Average Salary by Country
- Hiring Trend Analysis

Filters

- Country
 - Work Type
 - Role
 - Company Size
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Key Insights

- Identified countries with the highest number of job openings.
 - Analyzed work-type distribution across industries.
 - Identified the most common job roles.
 - Evaluated salary trends across countries.
 - Measured hiring trends over time.
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Conclusion

Successfully developed an end-to-end ETL pipeline using Python, Pandas, PostgreSQL, SQL, and Power BI. The project processed over 158,000 job records and generated valuable insights into global job market trends, salary distribution, hiring patterns, and work-type preferences.